

Experimental Activity 2 (EA2): Feedback control

Experimental activity EA2 regards feedback control, and it will be divided into two main parts, that are:

1. Trajectory tracking
2. Posture regulation

Consider as parameters the values that you identified in EA1, or the nominal values:

- Wheels radius: $r = 0.03 [m]$
- Wheels distance: $d = 0.165 [m]$

As regards saturation and sampling time, consider the following values:

- $\omega_M = 10 [rad/sec]$;
- $T_s = 0.04 [sec]$.

For localization, you can use one of the strategies seen so far (EKF or integration), with measures from:

- Encodes (wheels speed);
- Motion capture;

If you choose EKF, measures will be merged automatically by the algorithm. Instead, if you rely on simple integration strategies, we suggest setting q equal to the motion capture measure, if available, and use integration only when motion capture measures are not available.

Simulate trajectories to verify that they are feasible before testing them on the actual system. You can test the code without motion capture before performing the test with motion capture. If you keep the robot on the desk, **to prevent that the robot falls from the desk, keep the robot on its support platform, so that wheels run freely.**

You can deliver ONLY optional material in a single zip file. All the other exercises are propaedeutic to future activities. For this reason, it is important that you conclude them. If you encounter any issue, or results are not as you expected, please, discuss them with the professor or assistants.

PART 1: TRAJECTORY TRACKING

Test the three trajectory tracking control schemes seen in lectures and implemented in NA3, namely, the approximate linearization controller, the nonlinear feedback controller, or feedback linearization.

Activities:

- 1.1 Design a circular trajectory centered in $[0,0]$. Verify that wheel's speed modules necessary to follow the trajectory are smaller than the saturation value ω_M . Test the control schemes with the trajectory designed.
- 1.2 Plan a trajectory that brings the robot from the initial configuration $q_i = \left[-0.5, -0.5, \frac{\pi}{2}\right]$ to the final configuration $q_f = \left[0.5, 0.5, \frac{\pi}{2}\right]$ using cartesian polynomials or the chained form formulation. Choose as a time law a trapezoidal time law, such that modules of wheels speeds are lower than ω_M . Choose a controller and test it.

Optional material to deliver.

- Run 1.2, save results and write a script that plots figures useful to represent the desired and actual trajectory, and the tracking error. Name the script EX_opt_1.

PART 2: REGULATION

In these experiments, you have to test the regulation controllers seen in lectures.

Activities:

- 2.1 Test the two regulators setting the target configuration to the origin $(0,0,0)$. Starting from different initial configurations.
- 2.2 Test the two regulators setting the target to arbitrary configurations different from $(0,0,0)$.

Optional material to deliver.

- Run 2.2, save results and write a script that plots figures useful to represent the desired configuration to reach and the actual trajectory, as well as the tracking error. Name the script EX_opt_2.